

ML Assignment 4

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Section A (Theoretical)

Ques 1:

a.)	Input Dimensions : $M \times N$ channels : P Kernel size : $K \times K$
a)	Stride = 1 Feature Map Dimensions = (H, W) $= (M - K + 1, N - K + 1)$
b)	Total Multiplications = $K \cdot K \cdot P$ Additions = $K \cdot K \cdot P - 1$ Total Operations = $2 K \cdot K \cdot P - 1$
c)	Total operations = $(M - K + 1)(N - K + 1) \cdot Q \cdot P \cdot K \cdot K$ Also that $(M, N) \gg K$ $\Rightarrow (M + 1)(N + 1) \cdot Q \cdot P \cdot K \cdot K$ $\Rightarrow O(M \cdot N \cdot Q \cdot P \cdot K \cdot K)$

Ques 2:

Q.2) In the assignment step, we use fixed centroids and assign points to clusters according to distance.

$$\min \sum_{i=1}^n |x_i - \mu_{c_i}|^2$$

In the update step, we keep the assignment of points to clusters fixed as defined in assignment step. Then, we update the centroids based on the mean of the assigned points.

$$\mu_{c_i} = \frac{1}{|C_i|} \sum_{x \in C_i} x$$

We use elbow method to find the optimal number of clusters. In this, we run k-means algo for different number of clusters while calculating the within cluster sum of squares (WCSS). The point where the plot levels off is our optimal number of clusters.

No, random initialization of clusters does not guarantee ~~the~~ global minima. K-Means clustering is sensitive to initial coordinates of clusters. Random initialization doesn't guarantee global minima but it can give it sometimes.

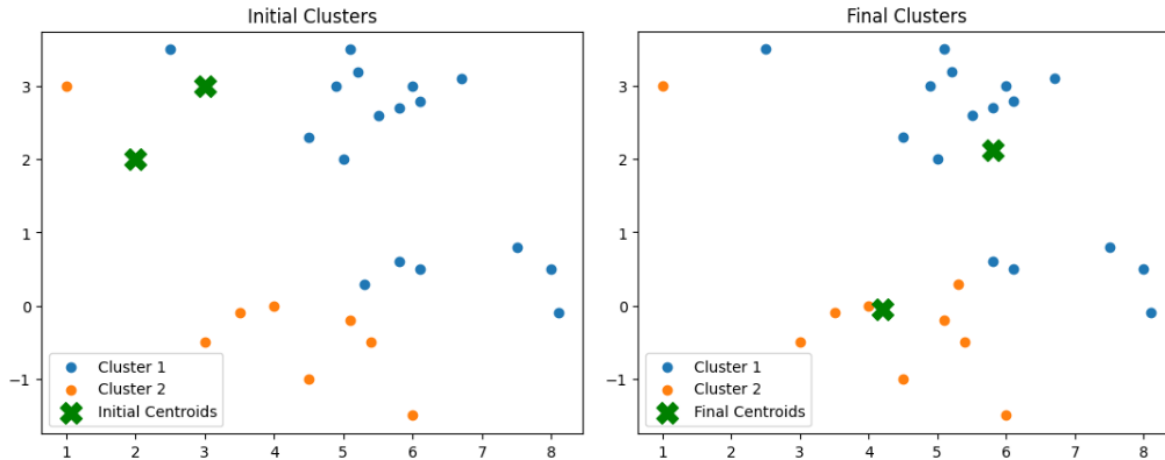
Section B (Scratch Implementation)

Part 2

The coordinates of the final centroids after the algorithm converges are as follows:

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Final Centroids: [[ 5.8      2.125 ]
 [ 4.2      -0.05555556]]
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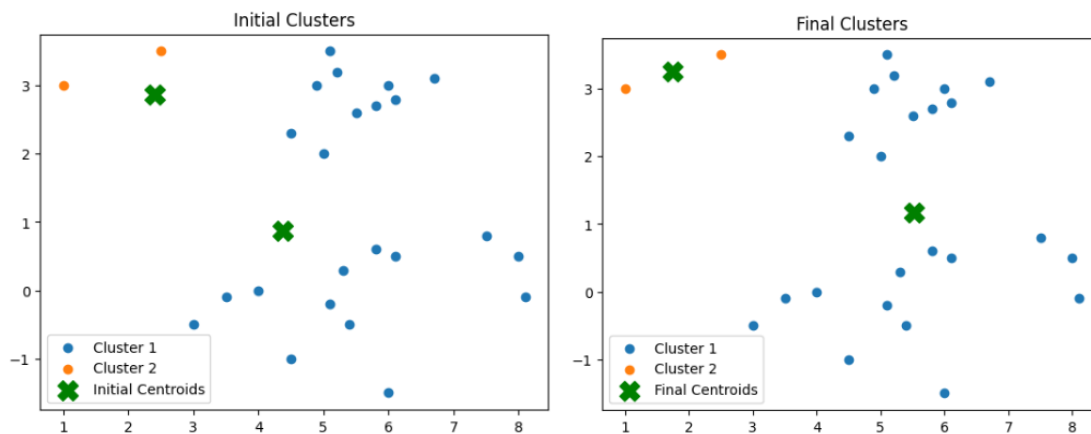
The initial and final clusters have been visualized in the plots below:



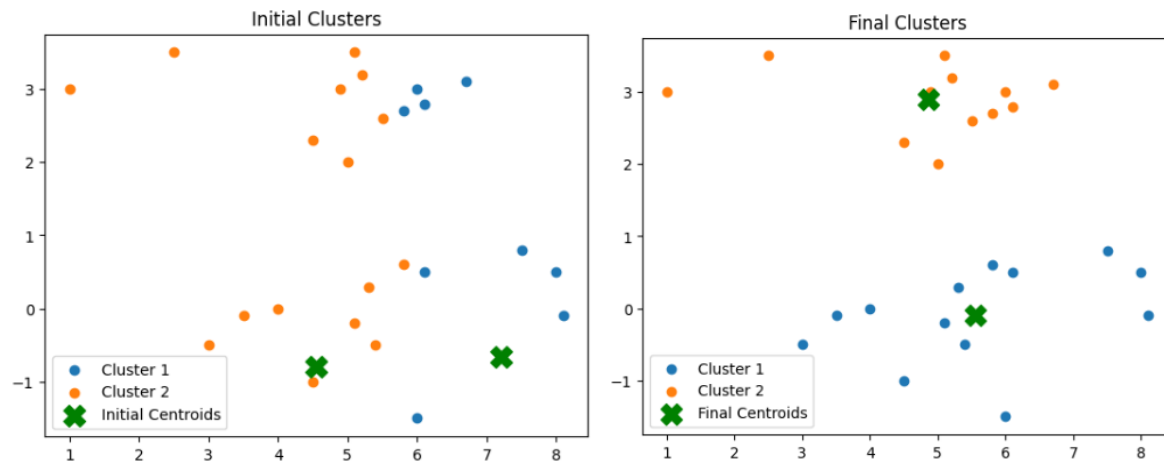
Part 3

The centroids were initialized twice. The initial and final cluster plots for both cases can be seen below.

Case 1:

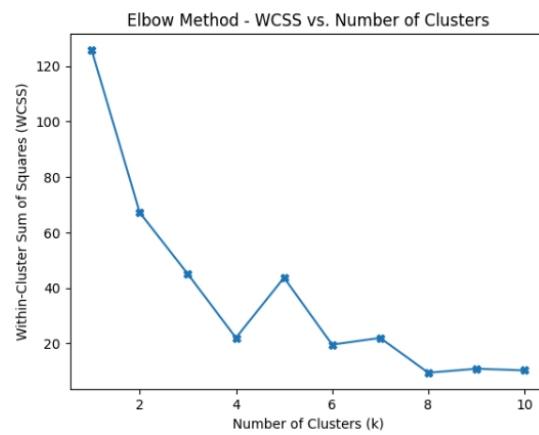


Case 2:



Part 4

The Elbow method plot can be seen below, indicating the optimal number of clusters is 4.



The final clusters after applying the K-means algorithm are shown below.

